

CRUISE: Vision-Language Model-Guided Uncertainty-Aware Cross-Modal Sensor Fusion for Robust Autonomous Driving

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Abstract—Modern autonomous vehicles are equipped with multiple sensors, such as cameras, LiDAR, and radar, for comprehensive environmental perception. However, robust cross-modal feature fusion remains a critical challenge, as the reliability of each sensor varies significantly across diverse real-world driving conditions, including poor visibility and adverse weather. While uncertainty quantification (UQ) mitigates this issue by allowing models to prioritize reliable signals, existing uncertainty-aware fusion methods typically rely on simple feature-level uncertainty estimates and thus often fail to generalize effectively in complex, out-of-distribution scenarios. To address this limitation, we propose CRUISE, a novel uncertainty-aware cross-modal sensor fusion framework. CRUISE integrates a vision-language model (VLM)-guided UQ module that generates fine-grained, pixel-level uncertainty estimates. By leveraging the VLM’s rich prior knowledge and superior contextual reasoning, our approach provides a highly informative guide for the fusion process. Furthermore, we introduce a dynamic adaptation mechanism that explicitly models and captures cross-modal dependencies, ensuring the framework fully exploits the inherent complementary nature of multi-sensor inputs. Our experiments show that CRUISE outperforms state-of-the-art (SOTA) methods by on average 4.87% in 3D object detection and 4.23% in semantic segmentation.

I. INTRODUCTION

Modern autonomous vehicles (AVs) employ sensor fusion to integrate data collected by diverse sensors, such as cameras, LiDAR, and radar, to achieve comprehensive perception of their surroundings [1], [2]. However, sensor reliability varies significantly across different scenarios, introducing uncertainty into perception and posing substantial challenges for robust feature fusion. For instance, while cameras capture rich visual details, they are susceptible to lighting variations and occlusions. Conversely, LiDAR and radar offer greater resilience against visual perturbations but are susceptible to multi-path interference and signal disruptions [3], [4], [5]. The dynamic nature of real-world driving means these reliability shifts are constantly present, demanding systems that can adapt on the fly. To address the challenge of unreliable sensor inputs, uncertainty quantification (UQ) is employed to estimate the confidence of each sensor modality. This mechanism enables fusion models to dynamically prioritize reliable information and suppress misleading signals [6], [7], [8], which is crucial for enhancing the safety and dependability of AV perception, particularly in safety-critical edge cases [6], [9], [10].

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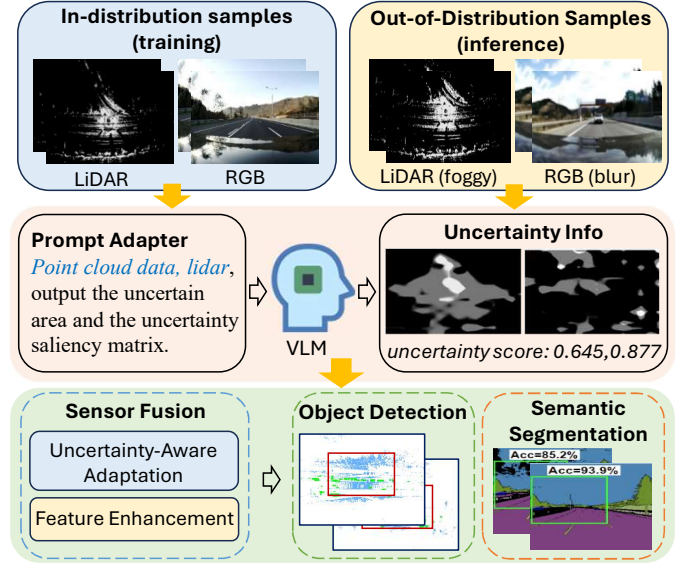


Fig. 1. Overview of CRUISE. A VLM-guided UQ module provides pixel-level uncertainty maps to guide sensor fusion, leading to more robust object detection and semantic segmentation.

Existing uncertainty-aware multi-modal sensor fusion typically adopt popular UQ techniques, such as Bayesian inference [11], [12], ensemble methods [13], [14], and Monte Carlo dropout, to improve robustness [15]. However, a fundamental challenge remains: these data-driven approaches often struggle to generalize effectively in complex, out-of-distribution (OOD) scenarios, where conditions differ significantly from those seen during training [16], [17], [18]. This limitation is especially critical for AVs, where encountering unseen conditions (e.g., severe weather or sudden sensor degradation) is inevitable in real-world deployments [1], [19]. A key contributing factor to this poor generalization is that most existing methods rely on coarse task-level or feature-level UQ, assigning a single uncertainty score to an entire modality or high-level representation. This results in an insufficient guidance signal. Consequently, they fail to capture the necessary fine-grained uncertainty variations at the pixel or voxel level, which are essential for dense prediction tasks like semantic segmentation and accurate 3D detection. Furthermore, these methods typically process each modality independently, often treating them as isolated data streams and lacking mechanisms to explicitly capture cross-modal dependencies [20], [21], [22], [23]. This deficiency prevents them from fully exploiting complementary sensor information, which is crucial when one sensor compensates for the weakness of another, leading to suboptimal representation learning and ineffective feature fusion.

To address these limitations, we propose CRUISE, a novel uncertainty-aware multi-modal sensor fusion framework that enhances the robustness and generalization of AV perception. Our approach is based on two key innovations. First, we leverage a Vision-Language Model (VLM) for fine-grained UQ. As shown in Figure 1, we adapt a pre-trained VLM (Anomaly GPT [24]), originally developed for industrial anomaly detection, to the autonomous driving domain via LoRA fine-tuning [25] on AV datasets. This adaptation allows the model to acquire not only domain-specific knowledge but also to reason over contextual relationships, which is vital for generalizing across various driving scenarios. During inference, the model first computes an uncertainty estimate by comparing the feature map of the test sample at each spatial location with representative training features. The VLM then refines this estimate by integrating local feature differences with global contextual reasoning, generating fine-grained, context-aware uncertainty heatmaps that highlight unreliable regions and guide the fusion process. Additionally, the VLM’s strong transferability improves generalization to OOD scenarios by leveraging prior knowledge and robust anomaly localization abilities. Second, to tackle the lack of cross-modal interaction, we further introduce a novel dynamic adaptation mechanism that explicitly captures cross-modal dependencies. This mechanism enables complementary sensor information (i.e., information where sensors compensate for each other’s weaknesses) to be adaptively balanced during fusion, thereby achieving a more optimal and robust final representation. Our main contributions are summarized as follows:

- We propose a CRUISE, a novel sensor fusion framework that incorporates a VLM-guided UQ module and a dynamic fusion mechanism to enhance multi-modal sensor fusion in AVs. Our approach achieves an average accuracy improvement of 4.87% in 3D object detection and 4.23% in semantic segmentation over SOTA approaches.
- To the best of our knowledge, this is the first VLM-based UQ method that leverages prior knowledge and contextual understanding to produce context-aware, pixel-level uncertainty maps. It significantly improves generalization to complex, out-of-distribution driving scenarios, outperforming SOTA UQ techniques by 4.16% on average.
- We design a novel fusion module that dynamically adjusts the contribution of each sensor with explicit modeling of cross-modal dependencies. This allows the model to exploit complementary information across modalities, improving robustness across diverse real-world scenarios.

II. RELATED WORK

A. Uncertainty Quantification

Various UQ techniques have been developed to enhance model robustness, including Bayesian neural networks that model weights as probability distributions [13], deep ensembles that estimate uncertainty by aggregating multiple predictions [11], and Monte Carlo (MC) dropout that introduces stochasticity during inference [15]. However, a critical limitation across these traditional methods is their failure to generalize effectively across environments, resulting in suboptimal

performance in complex or OOD scenarios [26]. Specifically, Bayesian models often suffer from poor scalability and rely on rigid probabilistic structures that fail to capture subtle contextual variations [13], [6], [27]. Ensemble methods tend to overfit the training distribution, lacking the capacity to extrapolate to unseen conditions [9], [28], [29]. While MC dropout introduces randomness to approximate uncertainty, it lacks the necessary adaptability in dynamic environments [7], [30]. In summary, these methods derive uncertainty largely from the statistical properties of the training data, making them inherently brittle when faced with distributional shifts. Recent advances in VLMs have demonstrated strong generalization capabilities across domains by leveraging large-scale pretraining [31], [32]. Unlike traditional models that rely solely on feature representations from a specific training distribution, VLMs incorporate extensive prior knowledge from vast and diverse datasets, enabling better adaptation to complex, OOD scenarios [33], [34], [35]. This ability to leverage cross-modal semantic understanding and prior knowledge offers a powerful alternative to purely data-driven uncertainty estimation. However, translating the strong conceptual understanding of VLMs into concrete, pixel-level uncertainty maps for low-level perception in multi-modal UQ remains largely unexplored.

B. Multi-modal Sensor Fusion in AVs

Multi-modal sensor fusion is fundamental to AV perception, integrating features from diverse sensors to construct a comprehensive representation of the environment [5], [36]. Leveraging complementary strengths of different modalities, sensor fusion enhances performance in complex, dynamic scenarios where single-modality data may be unreliable [37], [38], [39]. In 3D detection, frameworks such as BEVFusion [40] and TransFusion [41] align LiDAR point clouds and camera images via transformer-based architectures to capture both spatial depth and visual cues. For 2D segmentation, recent frameworks employ cross-modal attention [42] and hierarchical fusion layers [43] to adaptively weight sensor inputs [44], [45]. Yet, despite their architectural sophistication, these approaches remain vulnerable to disturbances and OOD conditions such as adverse weather or sensor noise [18]. To mitigate these risks, a second category of fusion approaches has emerged, explicitly focusing on robustness across driving scenarios. Designs such as HydraFusion [36] and EcoFusion [46] rely on coarse modality-level confidence, while more recent methods such as uncertainty-encoded fusion [7], Cocoon [47], SAMFusion [48], CAFuser [49], and ContextualFusion [50] introduce condition- or uncertainty-aware weighting to adapt sensor contributions under adverse conditions. However, despite these advances, most existing approaches focus on coarse, modality- or object-level uncertainty estimation, which is insufficient for dense prediction tasks where reliable fusion must occur at every spatial location. Furthermore, few methods explicitly model the uncertainty-driven interaction between modalities, which limits their ability to dynamically and precisely leverage complementary information. Our work addresses this dual gap by providing fine-grained, VLM-guided uncertainty and an explicit dynamic adaptation mechanism for fusion.

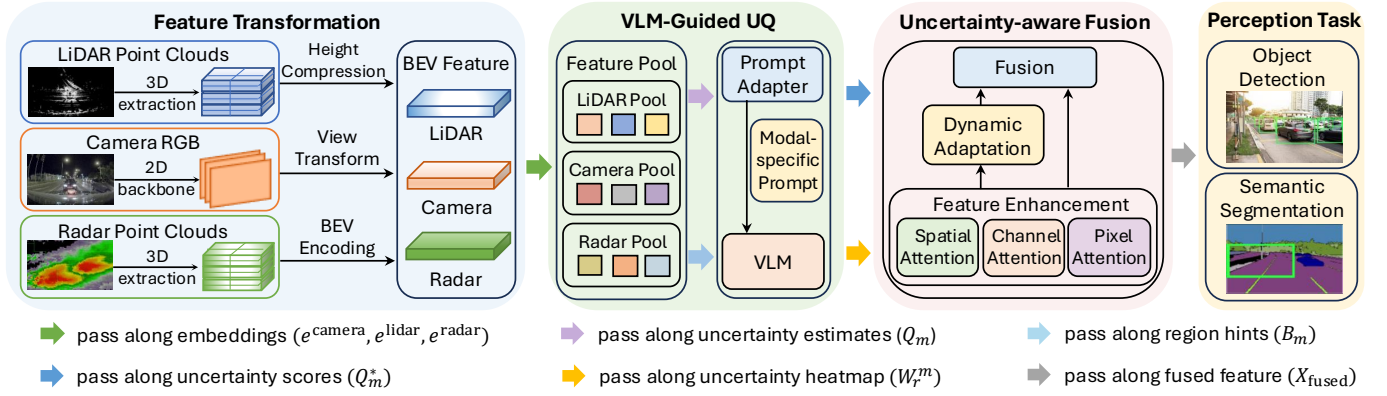


Fig. 2. 3D architecture of our uncertainty-aware multi-modal sensor fusion framework for AV perception tasks. The framework comprises three states: feature transformation into the BEV space; a VLM-guided UQ module that generates fine-grained uncertainty heatmaps using modality-specific prompts; and an uncertainty-aware fusion module that leverages a dynamic adaptation mechanism to produce fused features for object detection and semantic segmentation.

III. METHODOLOGY

A. Problem Formulation

We propose CRUISE, as demonstrated in Figure 2, a novel uncertainty-aware sensor fusion framework structured into two main stages: VLM-guided UQ and uncertainty-aware fusion. CRUISE first projects features captured by different modalities into a unified shared feature space, typically a Bird's-Eye-View (BEV) representation, to ensure feature alignment. A prompt adapter is then fine-tuned for a VLM (AnomalyGPT [24]) using these feature representations to allow the model to acquire essential prior knowledge specific to autonomous driving. During training, we construct a feature pool for each modality, storing features of representative training samples as a reference for uncertainty estimation. Subsequently, during inference, we compute an initial uncertainty estimate by comparing the test sample's feature map with these representative training samples in the feature pool. The VLM then refines these estimates through global contextual reasoning, leveraging its deep semantic understanding to generate a highly informative uncertainty heatmap that accurately localizes unreliable regions at a fine-grained spatial level. Notably, our model-agnostic UQ module can be integrated into existing fusion frameworks and extended to additional modalities via finetuning with corresponding data. In the uncertainty-aware fusion stage, CRUISE improves representation learning through spatial, channel, and pixel attention to emphasize critical regions, highlight informative features, and adjust the modality-specific importance. We further introduce a novel dynamic adaptation mechanism that explicitly models cross-modal dependencies, enabling modalities to reinforce reliable signals and compensate for uncertain features. Finally, CRUISE adaptively fuses features based on the fine-grained uncertainty information provided by the VLM-guided UQ, thereby effectively mitigating error propagation and significantly enhancing robustness in downstream perception tasks.

B. Multi-Modal Feature Extraction

We employ modality-specific backbones to extract feature representations from LiDAR point clouds, radar point clouds, and multi-view camera images. These heterogeneous features

are then projected into a shared feature space using learnable projection functions to ensure cross-modal alignment.

1) *LiDAR Features*:: We use VoxelNet [51] with SparseConvNet [52] to extract LiDAR features, encoding voxel-wise geometric properties into feature vectors that capture 3D spatial structures. The extracted feature for the i^{th} LiDAR sample is denoted as $\mathbf{f}_l^{(i)} \in \mathbb{R}^{C_l \times D_l \times H_l \times W_l}$, where C_l is the number of feature channels, and D_l , H_l , and W_l denote the depth, height, and width of the 3D feature map, respectively. We then apply height compression to obtain a 2D bird's eye view (BEV) representation $\mathbf{e}_l^{(i)} \in \mathbb{R}^{C \times H \times W}$, where C is the number of feature channels, H and W are the height and width of the shared feature space, respectively. This transformation preserves key geometric information while ensuring alignment with the shared feature space.

2) *Camera Features*:: We use ResNet-50 [53] to extract high-level image (RGB) features from multi-view camera images. The extracted feature for the i^{th} sample from the j^{th} view is denoted as $\mathbf{f}_c^{(i,j)} \in \mathbb{R}^{C_c \times H_c \times W_c}$, where C_c is the number of feature channels, and H_c and W_c are the height and width of the feature map, respectively. To ensure spatial alignment with other modalities, we apply view transformation to project image features into the BEV space, followed by a 1×1 convolution to map C_c channels to C channels. The final camera representation is $\mathbf{e}_c^{(i,j)} \in \mathbb{R}^{C \times H \times W}$, where C is the number of feature channels, H and W are the height and width of the shared feature space, respectively.

3) *Radar Features*:: We use PointPillars [54] to extract radar features by voxelizing radar point clouds into a sparse pillar-based BEV representation. Each pillar aggregates radar reflections, capturing both spatial distribution and Doppler velocity information. The extracted representation for the i^{th} radar sample is denoted as $\mathbf{f}_r^{(i)} \in \mathbb{R}^{C_r \times H_r \times W_r}$, where C_r is the number of feature channels, and H_r , W_r are the height, and width of the BEV feature map. The radar features are then processed using a SparseCNN [55] to extract global spatial features while preserving motion information. Finally, a 1×1 convolution is applied to ensure feature compatibility, producing the final radar feature representation as $\mathbf{e}_r^{(i)} \in \mathbb{R}^{C \times H \times W}$, where C , H , W are the number of feature channels, height, and width of the shared feature space.

C. Feature Pool

1) *Core Sampling*: Each training sample comprises modality-specific inputs from camera, radar and LiDAR with feature representations $\mathbf{e}_i^{\text{camera}}, \mathbf{e}_i^{\text{LiDAR}}, \mathbf{e}_i^{\text{radar}} \in \mathbb{R}^{C \times H \times W}$ ($1 \leq i \leq N$), where N is the number of training samples, and C, H, W denote the number of feature channels, height, and width of the shared feature space, respectively. Note that these feature representations retain their spatial structure rather than being flattened vectors. To reduce memory and computational cost, we apply core sampling [56] to construct feature pools $\mathcal{P}_{\text{camera}}, \mathcal{P}_{\text{LiDAR}}, \mathcal{P}_{\text{radar}} \in \mathbb{R}^{N' \times C \times H \times W}$, selecting N' ($N' \ll N$) representative features for each modality. Denoting the full set of feature representations of each modality as $\mathcal{E}_m \in \mathbb{R}^{N \times C \times H \times W}$, where $m \in \{\text{camera, LiDAR, radar}\}$, we define the closest representative $\mathbf{p} \in \mathcal{P}_m$ for each $\mathbf{e} \in \mathcal{E}_m$ using the Euclidean distance:

$$d(\mathbf{e}, \mathcal{P}_m) = \min_{\mathbf{p} \in \mathcal{P}_m} \sum_{i=1}^H \sum_{j=1}^W \|\mathbf{e}_{:,i,j} - \mathbf{p}_{:,i,j}\|_2. \quad (1)$$

We iteratively add the feature representation with the largest distance d to the pool until it reaches N' samples. By minimizing the worst-case distance between features in $\mathcal{E}_m$ and their nearest representatives in $\mathcal{P}_m$, the feature pool efficiently preserves the diversity of training samples. At inference, we estimate uncertainty at each spatial location in the test sample's feature representation by comparing it to the embeddings at the same location in the representations stored in the feature pool of the corresponding modality.

2) *Uncertainty Estimation*: At inference, we compute the uncertainty estimates $Q_m \in \mathbb{R}^{H \times W}$ for a test sample $\mathbf{v} \in \mathbb{R}^{C \times H \times W}$ of modality m . Each element Q_{ij}^m ($0 \leq i \leq H, 0 \leq j \leq W$) in Q_m , represents the minimum Euclidean distance between the feature vector $\mathbf{v}_{ij} \in \mathbb{R}^C$ at spatial location (i, j) in the feature representation of the test sample and its closest feature vector $\mathbf{p}_{ij} \in \mathbb{R}^C$ at the same location of all the representative features in the feature pool, i.e.,

$$Q_{ij}^m = \min_{\mathbf{p} \in \mathcal{P}_m} \|\mathbf{v}_{ij} - \mathbf{p}_{ij}\|_2. \quad (2)$$

A higher Q_{ij}^m value indicates greater uncertainty at location (i, j) . Additionally, we introduce region hints $B_m \in \mathbb{R}^{H \times W}$ by thresholding Q_m , highlighting high-uncertainty regions in the feature representation. The uncertainty estimates and region hints are then passed to the VLM, integrating local uncertainty estimates with contextual reasoning.

D. VLM-Guided Uncertainty Quantification

We fine-tune AnomalyGPT [24] using LoRA [25] in an unsupervised manner to align visual uncertainty cues (i.e., localization map M) with language-based reasoning (i.e., uncertainty map A_{text}). AnomalyGPT generates uncertainty heatmaps that localize abnormal regions based on both low-level details and high-level semantic context. We use AnomalyGPT for its dual-stream design and strong localization, but our method can be generalized to any VLM supporting vision-text alignment and region-level output. AnomalyGPT is designed to produce uncertainty heatmaps that localize

abnormal regions in input images. It employs convolutional front-end layers to capture low-level details, followed by attention mechanisms for high-level semantic reasoning.

1) *Visual Stream*: The input feature representations to the VLM are, $\mathbf{e}_i^{\text{camera}}, \mathbf{e}_i^{\text{LiDAR}}, \mathbf{e}_i^{\text{radar}} \in \mathbb{R}^{C \times H \times W}$, where i indexes the sample, C is the number of channels, and H and W are the height and width of the shared feature space, respectively. These features are processed by a frozen ImageBind encoder [57], which extracts multi-scale patch-level features from hierarchical layers, generating feature maps F_j^{patch} , where j indexes the hierarchical layer depth. These features are then passed through a lightweight transformer-based decoder [24] that aligns visual features with learned textual semantics. The decoder performs visual-textual feature matching and outputs a fine-grained uncertainty localization map $M \in \mathbb{R}^{H \times W}$. Specifically, it aligns the intermediate patch-level features F_j^{patch} with learned textual features F_{text} , which capture semantics of normal and abnormal scenarios, enabling fine-grained uncertainty localization and quantification across diverse environments. The decoder computes similarity scores between patch-level features and textual features, then applies up-sampling to generate a high-resolution localization map $M \in \mathbb{R}^{H \times W}$, indicating uncertainty regions in the feature.

2) *Textual Stream*: Powered by Vicuna-7B [58], the textual stream interprets uncertainty via attention-based reasoning using four inputs: (i) a localization map M from the visual stream, (ii) a user query q_{user} (e.g., "Describe the detected uncertainty in this image"), (iii) a structured prompt E_{prompt} from the prompt adapter (details below), and (iv) feature representations. These inputs are processed by Vicuna-7B, generating an attention-based uncertainty map A_{text} from cross-modal attention scores over E_{region} . This map highlights uncertain regions, and a context-aware pixel-level uncertainty scores $Q_m^* \in \mathbb{R}^{H \times W}$ is obtained by interpolating attention scores to match the resolution of the visual stream.

• **Prompt Adapter**: To bridge the vision and textual streams, we introduce a prompt adapter that converts uncertainty estimates Q_m , region hints B_m ($m \in \{\text{camera, LiDAR, radar}\}$), and localization maps M into structured prompts. The structured prompt is formed by concatenating a base prompt E_{base} , an uncertainty embedding $E_{\text{uncertain}}$, and refined region hints E_{region} . Specifically, the base prompt E_{base} provides uncertainty detection objective and modality specification (e.g., camera, LiDAR, radar). The uncertainty embedding $E_{\text{uncertain}}$ is derived from the uncertainty estimates Q_m and compressed with a CNN [59] to ensure compatibility. The refined region hints E_{region} is constructed from the localization map M and the region hints B_m with positional encoding to emphasize high-uncertainty regions. The final structured prompt is defined as $E_{\text{prompt}} = [E_{\text{base}}, E_{\text{uncertain}}, E_{\text{region}}]$.

3) *Integration of Visual and Textual Streams*: We integrate the outputs from both the visual and textual streams to produce the final uncertainty heatmap W_r^m that precisely localizes uncertainty areas in the original images:

$$W_r^m = \lambda M + \gamma A_{\text{text}} \quad (3)$$

where M is the visually derived localization map, A_{text} is the contextually derived uncertainty map, and λ and γ are hyperparameters that control the relative contributions of the visual and textual information, respectively. This final heatmap W_r^m provides the refined, context-aware guidance signal used in the subsequent uncertainty-aware fusion stage.

E. Uncertainty-aware Sensor Fusion

We propose a novel uncertainty-aware fusion module that adaptively fuses features from each sensor based on the UQ information provided by the VLM to enhance robustness. We first use the uncertainty heatmap W_r^m as a mask to suppress unreliable regions, generating a refined representation

$$X'_i = X_i \odot (1 - W_r^m) \quad (4)$$

where $\odot$ denotes element-wise multiplication and X_i is the initial feature representation of modality m . We reduce uncertainty propagation by assigning smaller weights to high-uncertainty regions in W_r^m .

1) **Feature Enhancement:** We employ a combination of spatial, channel, and pixel-level attention mechanisms to capture comprehensive contextual information.

- **Spatial Attention:** The spatial attention mechanism captures spatial dependencies by generating a weight map S_i for each modality's feature map X'_i , highlighting high-confidence (low-uncertainty) regions. We apply global average pooling X'_{avg} and max pooling X'_{max} across the channels of X'_i , concatenate the descriptors, and process them through a convolutional layer to refine the attention weights:

$$S_i = \sigma(\text{Conv2D}(\text{Concat}(X'_{\text{avg}}, X'_{\text{max}}))), \quad (5)$$

where σ denotes the sigmoid activation function.

- **Channel Attention:** The channel attention mechanism enhances informative channels by dynamically adjusting their importance. We apply global average pooling on X'_i to obtain a channel descriptor T_i , which is then passed through a two-layer multi-layer perceptron to generate channel-wise attention weights T'_i . These weights emphasize significant channels while suppressing less useful ones. The enhanced feature representation is computed as:

$$X''_i = X'_i \odot S_i \odot T'_i, \quad (6)$$

where $\odot$ denotes element-wise multiplication. The resulting feature map X''_i effectively integrates spatial- and channel-wise significant features, emphasizing relevant information while suppressing noisy or unreliable regions.

- **Pixel Attention:** The pixel attention mechanism facilitates cross-modal fusion by adaptively weighting each modality at the pixel level. We concatenate features from different modalities and process them through a convolutional layer to produce a pixel-wise attention map G_i :

$$G_i = \sigma(\text{Conv2D}(\text{Concat}(X''_i, X''_0))) \quad (7)$$

where X''_0 denotes the feature map of another modality that has also undergone spatial and channel attention. The final fused representation is computed as

$$X_{\text{final}} = G_i \odot X''_0 + (1 - G_i) \odot X''_i \quad (8)$$

where $\odot$ denotes element-wise multiplication. This mechanism adaptively adjusts the contribution of each modality at the pixel level, emphasizing high-confidence regions while suppressing unreliable information. We extend Eq. (7)(8) to three modalities fusion by sequentially applying the same fusion step twice: first fusing two modalities (e.g., LiDAR + camera), then fusing the result with the third (e.g., radar).

2) **Modality Dynamic Adaptation:** We introduce a novel dynamic adaptation mechanism that adaptively adjusts the weight of each modality in feature fusion based on the uncertainty score Q'_m , a normalized scalar in $[0, 1]$ obtained by aggregating pixel-wise uncertainty scores Q_m^* . This mechanism ensures that more reliable modalities receive higher weights and contribute more to the fused representation. The weighting function achieves this by incorporating both uncertainty scores and inter-modality variance. Specifically, the weight λ_m for each modality $m \in \{\text{camera, LiDAR, radar}\}$ is defined as:

$$\lambda_m = \frac{\exp(-\nu \cdot Q'_m) + \gamma \cdot (\frac{1}{M} \sum_m (Q'_m - \bar{Q})^2)}{\sum_m (\exp(-\nu \cdot Q'_m) + \gamma \cdot (\frac{1}{M} \sum_m (Q'_m - \bar{Q})^2))}, \quad (9)$$

where M is the total number of modalities, $\bar{Q}$ is the mean uncertainty score across all the modalities, ν controls sensitivity to the uncertainty score, and γ adjusts sensitivity for inter-modality variance. By dynamically weighting modalities based on their uncertainty and inter-modality variance, this adaptive weighting enhances feature fusion by explicitly incorporating cross-modal dependencies. The final fused feature X_{fused} is then computed as a weighted sum of each modality's feature representation X_{final}^m :

$$X_{\text{fused}} = \sum_m \lambda_m X_{\text{final}}^m \quad (10)$$

X_{fused} is then passed to a task head to perform downstream perception tasks. The task-specific loss is denoted as L_{baseline} .

3) **Modality Consistency Loss:** We introduce a UQ loss and a modality consistency loss to reduce inconsistencies across modalities and stabilize representations in uncertain regions.

- **UQ Loss:** The UQ loss $L_{\text{uncertainty}}$ penalizes the differences between feature representations in identified high-uncertainty regions e_r^m and their closest representative features e_l^m from the feature pool $\mathcal{P}_m$, where $m \in \{\text{camera, LiDAR, radar}\}$. By minimizing this loss, the model stabilizes feature representations in uncertain areas and prevents significant deviation from the representative features in the feature pool. This promotes feature consistency and reduces uncertainty propagation, improving the robustness of downstream tasks. Mathematically,

$$L_{\text{uncertainty}} = \sum_m \sum_r \mathbb{I}(W_r^m > \tau) \cdot \|e_r^m - e_l^m\|_2, \quad (11)$$

where $\mathbb{I}$ is an indicator function that activates for each pixel (i, j) in the heatmap W_r^m that exceeds a threshold τ .

- **Modality Consistency Loss:** The modality consistency loss $L_{\text{consistency}}$ enforces feature alignment across modalities in high confidence (low-uncertainty) regions, ensuring coherent multi-modal fusion. Mathematically,

$$L_{\text{consistency}} = \sum_{m_1 \neq m_2} \sum_{r \in \mathbb{I}(W_r^m \leq \tau)} \|e_r^{m_1} - e_r^{m_2}\|_2, \quad (12)$$

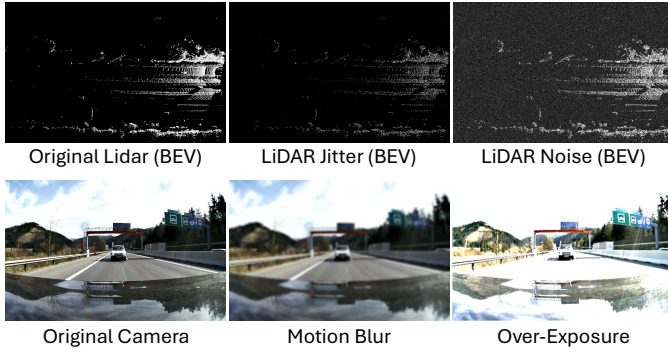


Fig. 3. Original and Simulated Out-of-Distribution Samples: The figure displays LiDAR data perturbed by jitter and noise, and camera images degraded by motion blur and over-exposure. These samples, alongside the original inputs, are utilized for robust evaluation under OOD conditions.

where m_1 and m_2 denote different modalities, τ denote a predefined threshold, and r denotes the uncertain regions. The total loss function is then defined as:

$$L_{\text{total}} = L_{\text{baseline}} + \alpha L_{\text{uncertainty}} + \beta L_{\text{consistency}}, \quad (13)$$

where α and β are coefficients adjusting the importance of UQ loss and modality consistency loss, respectively.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

Datasets: We evaluate CRUISE under various adverse conditions using three public autonomous driving datasets.

- **aiMotive** [60]: The dataset consists of 26,583 annotated frames of camera and LiDAR data for 3D object detection, covering over 425,000 objects across 14 classes. It spans diverse driving conditions (highway, night, rain, and urban) and is split into 21,402 training frames and 5,181 validation frames. We simulated 5181 out-of-distribution (OOD) scenarios for inference by applying motion blur (MB), over-exposure (OE), under-exposure (UE), and LiDAR noise (LN).
- **DeLiVER** [61]: The dataset consists of 47,310 samples, including 1406 OOD samples, with six views and four sensor modalities (cameras, LiDAR, event, and depth sensors) for semantic segmentation under challenging conditions (cloudy, foggy, night, rainy). Our OOD simulation includes MB, OE, UE, LN, and event low-resolution.
- **MFNet** [62]: This dataset consists of 1,569 paired visual (RGB) and thermal infrared images, including 820 daytime and 749 nighttime samples. We simulated 392 OOD samples for inference.

Evaluation Metrics: Following SOTA research, we evaluate CRUISE on the aiMotive dataset using all-point/11-point interpolation average precision (AP) metrics for object detection, and on DeLiVER and MFNet using mean Intersection over Union (mIoU) for semantic segmentation.

B. Implementation Details

1) *Model Specification:* We implemented CRUISE with PyTorch on an Ubuntu system with four NVIDIA RTX 3090 GPUs. The VLM was fine-tuned with abnormal data samples

(e.g., night, snow) from the aiMotive [60], DeLiVER [61] (excluding OOD samples), and MFNet [62]. Since our workflow is 3D-based, but DeLiVER and MFNet contain 2D images, we performed a 3D-to-2D migration by extracting features from the 2D datasets (e.g., DeLiVER and MFNet) using ResNet-50 [53] to build a feature pool for alignment with our framework. We train our model on aiMotive with the AdamW optimizer [80] with the learning rate of 0.0000625 and weight decay at 10^{-6} , on DeLiVER using the AdamW optimizer [80] with the learning rate of 0.00006 and weight decay at 0.01, and on MFNet [62] with the momentum SGD [81] with the learning rate of 0.01 and decayed by a factor of 0.94 per epoch. Each baseline was trained with a batch size of 1 for 200 epochs or until convergence. We set the parameters ν to 1 and γ to 0.5 in the weighting function (equation (9)), and both parameters α and β to 1 in the loss function (equation (13)). We set the threshold τ in the UQ loss function (equation (11), equation (12)) to 0.5. Our method is compared against SOTA UQ techniques and multi-modal fusion works, following each method's open-source implementation and hyperparameter settings.

2) *OOD Sample Generation:* Each model is trained on in-distribution (ID) training data and evaluated on simulated OOD samples to assess generalization and robustness under high-uncertainty conditions. We simulated OOD samples for aiMotive and MFNet, as DeLiVER inherently includes a separate OOD set. Camera OOD samples are generated via image transformations such as Gaussian blur and exposure shifts [82], [83]. For LiDAR, we apply random point dropping, noise injection, and partial range removal to mimic incomplete sampling, sensor errors, and occlusions [18]. For radar in aiMotive, we convert JSON-formatted data into point clouds and apply the same perturbations. Examples of our simulated OOD samples are provided in Fig. 3.

3) *VLM Fine-tuning:* We fine-tune the VLM [24] using LoRA with parameters set to $\text{lora}_r = 32$, $\text{lora}_\alpha = 32$, and $\text{lora}_{\text{dropout}} = 0.1$ for efficient parameter adaptation. The vision stream employs a frozen ImageBind-Huge [57] model, extracting patch-level features from layers 8, 16, 24, and 32. Apart from the incorporation of a prompt adapter, the base AnomalyGPT model [24] remains largely unchanged, retaining its pretrained weights and architectural configurations. Fine-tuning is conducted using the AdamW optimizer [80] with a learning rate of $1e^{-3}$ and a one-cycle cosine learning rate schedule for stable convergence. For the uncertainty heatmap (equation (3)), we set $\lambda = 1$ and $\gamma = 0.5$ to balance the contributions from the visual and textual streams.

C. Experimental Results

1) *Multi-tasks Perception Comparison:* We compare CRUISE against SOTA multi-modality fusion methods in Tables I, II, and III. As shown in Table I, on aiMotive, CRUISE delivers on average 3.82% better performance in all-point AP and 3.56% in 11-point interpolation AP compared to its baseline (aiMotive) without UQ. It also consistently outperforms task-specific SOTA sensor fusion methods, e.g., BEVFusion [40] and MSMD Fusion [66]. Note that our model-agnostic UQ module can be integrated into other SOTA

Scenarios	Sensor Fusion Method				Baseline		Ours	
	FocalFormer3D [63]	MVP [64]	BEVFusion [40]	CMT [65]	MSMDFusion [66]	aiMotive [67]	CRUISE	
Highway	69.94/67.72	68.74/66.60	76.89/74.64	73.50/70.98	75.64/73.56	75.67/73.50	76.27/74.09	(+0.60/+0.59)
Urban	56.96/56.16	59.71/58.49	58.37/57.70	60.18/59.01	61.77/60.49	61.10/60.01	63.29/62.23	(+2.19/+2.22)
Night	66.42/64.75	70.12/68.15	73.08/71.54	71.22/69.38	74.21/72.08	73.44/71.21	75.96/72.56	(+2.52/+1.35)
Rain	40.42/40.19	42.85/41.75	44.92/43.87	43.01/41.75	45.01/43.92	41.18/40.21	44.17/43.14	(+2.99/+2.93)
Mean	61.21/59.95	63.36/61.87	67.73/ 66.06	65.10/63.80	67.76 /65.96	65.85/64.46	68.37/66.76	(+2.52/+2.30)

TABLE I

MULTI-SENSOR FUSION PERFORMANCE ON THE aiMOTIVE DATASET. METRICS ARE REPORTED AS ALL-POINT AP/11-POINT INTERPOLATION AP (%). THE BEST (SECOND-BEST) RESULTS ARE HIGHLIGHTED IN PINK (PURPLE). IMPROVEMENTS OVER THE BASELINE ARE SHOWN IN BLUE.

Scenarios	Sensor Fusion Method				Baseline	Ours
	StitchFusion [68]	CMX [69]	TokenFusion [43]	HRFuser [70]	CMNeXt [61]	CRUISE
Cloudy	50.63 /55.89	48.96/56.00	43.10/49.19	37.56/39.69	50.48/ 56.61	53.93/58.25 (+3.45/+1.64)
Foggy	50.65 /55.12	49.93/54.24	43.57/48.88	35.10/40.53	50.58/ 55.13	54.27/57.94 (+3.66/+2.81)
Night	44.65/ 49.21	42.75/47.47	40.13/41.04	30.99/33.49	44.95 /48.02	47.19/49.72 (+2.24/+1.70)
Rainy	52.09/56.33	51.39/55.68	45.69/49.02	39.40/41.12	51.34/55.05	55.52/58.03 (+4.18/+2.98)
Sunny	50.30 /57.35	49.15/ 57.87	45.14/51.08	38.41/42.17	49.24/57.14	52.25/58.01 (+3.01/+0.87)
Mean	49.67/54.88	48.44/54.25	43.53/47.84	36.29/39.40	49.32/54.39	52.63/56.79 (+3.31/+2.40)

TABLE II

MULTI-MODAL SENSOR FUSION FOR RGB-LIDAR ON DeLiVER TEST SET (OUT-OF-DISTRIBUTION DATA)/VALIDATION SET (IN-DISTRIBUTION DATA) UNDER ADVERSE WEATHER CONDITIONS (METRICS: mIoU). THE BEST (SECOND-BEST) RESULTS ARE HIGHLIGHTED IN PINK (PURPLE). IMPROVEMENTS OVER THE BASELINE ARE INDICATED IN BLUE.

Method	Modality	UQ Method	mIoU (%)	Inference Latency
SwinT [71]	RGB	$\times$	40.23	—
SegFormer [72]	RGB	$\times$	43.21	—
ACNet [73]	RGB-Thermal	$\times$	44.80	—
FuseSeg [74]	RGB-Thermal	$\times$	51.44	—
FEANet [75]	RGB-Thermal	$\times$	53.49	—
SegMiF [76]	RGB-Thermal	$\times$	53.98	—
CMX (B4) [69]	RGB-Thermal	$\times$	57.14	—
CMNeXt [61]	RGB-Thermal	$\times$	58.47	—
CMNeXt [61]	RGB-Thermal	InfMCD [77]	58.62	0.56s
CMNeXt [61]	RGB-Thermal	InfNoise [77]	58.86	0.59s
CMNeXt [61]	RGB-Thermal	LDU [78]	58.58	0.50s
CMNeXt [61]	RGB-Thermal	CRUISE(ours)	59.78	0.66s

TABLE III

MULTI-SENSOR FUSION PERFORMANCE ON MFNet [62]. THE BEST (SECOND-BEST) RESULTS ARE HIGHLIGHTED IN PINK (PURPLE).

fusion frameworks to enhance performance. As shown in Table II, CRUISE exhibits better performance compared to SOTA sensor fusion frameworks across all the scenarios on DeLiVER, achieving on average 6.71% higher mIoU on the test set and 4.41% higher mIoU on the validation set over its baseline (CMNeXt) without UQ. It also outperforms other SOTA fusion framework by on average 5.95% higher mIoU on the validation dataset and 3.48% higher mIoU on the test dataset. On the MFNet (Table III), CRUISE delivers on average 2.24% higher mIoU over the second-best result. These results demonstrate CRUISE's better generalization in various OOD scenarios over SOTA approaches.

2) *UQ Methods Comparison*: Tables III, IV and V present CRUISE's performance and inference latency across various OOD scenarios on MFNet, aiMotive and DeLiVER, respec-

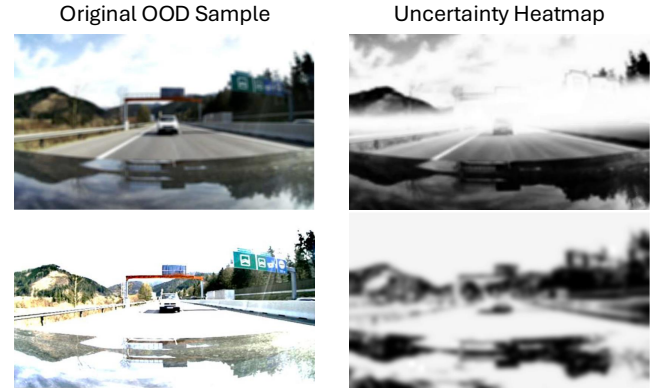


Fig. 4. Visualization of uncertainty heatmaps (W_r^m) generated by the VLM-guided UQ module for OOD samples. The figure shows uncertainty heatmaps for two camera OOD samples (motion blur and over-exposure). Brighter areas in the heatmap correspond to high-uncertainty regions, effectively localizing the visually degraded parts of the image.

tively. CRUISE outperforms the second-best model by on average 2.98% in all-point AP and 2.20% in 11-point interpolation AP on aiMotive, 1.56% on MFNet, and 4.24% higher mIoU on the testing set and 4.08% higher mIoU on the validation set of DeLiVER. These results demonstrate that our approach provides effective UQ with minimal additional latency.

3) *Qualitative Result*: Figure 4 demonstrates the uncertainty heatmaps produced by the VLM module for out-of-distribution (OOD) inputs. The results show that the VLM assigns higher uncertainty to regions affected by degradation (e.g., fog or blur), while maintaining lower uncertainty in well-structured areas. Such fine-grained uncertainty estimation enables the fusion module to down-weight unreliable signals and improve the robustness of downstream perception tasks.

UQ Method	Motion-Blur	Over-Exposure	Under-Exposure	LiDar-Noise	Mean	Inference Latency
PostNet [79]	60.42/58.90	63.12/59.99	60.20/59.25	64.13/60.60	61.97/59.69	—
InfMCD [77]	62.00/61.99	64.47/64.28	63.25/63.49	64.27/63.81	63.50/ 63.39	0.60s
InfNoise [77]	62.61/62.61	64.94/ 64.72	63.18/60.07	65.19/ 64.99	63.98 /63.10	0.62s
LDU [78]	62.06/62.22	65.07 / 64.85	62.87/60.02	65.47 / 65.10	63.87/63.05	0.55s
CRUISE (ours)	66.02/64.78	65.58 /64.29	66.01/64.78	65.96 /64.62	65.89/64.62	0.70s

TABLE IV

BEVDEPTH BASED UQ RESULTS ON aiMOTIVE UNDER CORNER CASES: MOTION-BLUR, OVER-EXPOSURE, UNDER-EXPOSURE, LiDAR-NOISE. METRICS: ALL-POINT AP / 11-POINT INTERPOLATION AP (%). THE BEST (SECOND-BEST) RESULTS ARE HIGHLIGHTED IN PINK (PURPLE).

UQ Method	Motion-Blur	Over-Exposed	Under-Exposed	LiDAR-Jitter	Event Low-Resolution	Mean	Inference Latency
PostNet [79]	42.82/53.91	43.77/51.74	29.98/33.13	47.95/52.44	48.08/56.23	42.41/47.86	—
InfMCD [79]	45.29/56.25	44.53/52.85	30.14/ 36.06	48.76/54.06	49.50/57.30	43.64/51.30	0.58s
InfNoise [77]	47.31/56.93	44.66/52.71	31.73 /34.81	48.79/ 54.79	50.01/ 57.97	44.50/51.44	0.53s
LDU [78]	46.01/56.35	45.00/52.96	30.38/33.46	48.98 /53.19	50.47 /57.95	44.17/50.78	0.47s
CRUISE (ours)	49.94/58.06	46.68/55.32	33.97/37.58	50.83/56.19	52.68/59.56	46.39/53.54	0.64s

TABLE V

CMNEXt BASED UQ RESULTS ON DeLiVER TEST SET (OUT-OF-DISTRIBUTION DATA)/VALIDATION SET (IN-DISTRIBUTION DATA) WITH CORNER CASES: MOTION-BLUR, OVER-EXPOSURE, UNDER-EXPOSURE, LiDAR-JITTER, AND EVENT LOW-RESOLUTION. METRICS: mIoU (%). THE BEST (SECOND-BEST) RESULTS ARE HIGHLIGHTED IN PINK (PURPLE).

D. Ablation Study

1) *Core Sampling Algorithm*: As demonstrated in Table VI, we compare the performance and inference latency of our core sampling algorithm with using other sampling algorithms [84], [85], [86] for constructing the feature pool. Across all datasets, CRUISE outperforms detrimental point processes (DPP) by 0.28% while improving inference latency by 7.6%. It also outperforms vector quantised-variational autoencoder (VQ-VAE) method by 3.67% with minimal increase in latency overhead. These results demonstrate that our core sampling approach effectively preserves the diversity of training samples while maintaining computational efficiency.

Method	aiMotive [67] (all-point AP/latency)	DeLiVER [61] (mIoU/latency)	MFNet [62]
VQ-VAE [84]	63.15/ 0.64s	44.08/ 0.61s	57.96/ 0.63s
KMC [86]	64.76/ 0.65s	45.46 / 0.61s	59.13/ 0.64s
DPP [85]	65.73 /0.76s	46.25/0.69s	59.57 /0.69s
CRUISE (ours)	65.89 /0.70s	46.39 / 0.64s	59.78 /0.66s

TABLE VI

COMPARISON OF DIFFERENT CORE SAMPLING ALGORITHM.

2) *VLM-Guided Uncertainty Quantification*: Table VII clearly demonstrates the substantial benefits of incorporating the VLM and the prompt adapter within the CRUISE framework. The baseline comparison, the “w/o VLM” variant, is constructed to isolate the VLM’s contribution: it retains the full downstream feature enhancement and dynamic adaptation pipeline, but critically replaces the refined VLM-guided uncertainty heatmap W^m with the raw, initial uncertainty estimate Q_m . This controlled setup ensures a fair comparison by rigorously evaluating the quality of the UQ signal alone. Across all datasets and perception tasks, integrating the VLM leads to a substantial and consistent performance improvement, with an average increase of 3.42% compared to the variant without it. This result emphatically highlights the VLM’s pivotal role in transforming simple distance-based

estimates (Q_m) into context-aware, fine-grained uncertainty maps (W^m), thereby significantly enhancing uncertainty estimation and subsequent feature fusion. Furthermore, the act of explicitly conditioning the VLM using the Prompt Adapter further improves performance by an additional 0.81%. This incremental gain validates the Prompt Adapter’s effectiveness in refining the VLM’s uncertainty reasoning and providing the necessary contextual and positional understanding to the textual stream. Notably, a key advantage of our design is its computational efficiency: these performance enhancements come at a minimal computational cost, as the overall inference latency increases by only a marginal 0.01s to 0.02s.

Methods	CRUISE w/o VLM	CRUISE w/o prompt adaptor	CRUISE w prompt adaptor
aiMotive [67]	63.56/ 0.67s	65.28 / 0.69s	65.89 /0.70s
DELIVER [61]	44.52/ 0.61s	46.08 / 0.63s	46.39 /0.64s
MFNet [62]	58.66/ 0.63s	59.27 / 0.64s	59.78 /0.66s

TABLE VII

COMPARISON OF THE EFFECTIVENESS OF OUR VLM AND PROMPT ADAPTER (METRICS: mIoU/INFERENCE LATENCY FOR DeLiVER AND MFNET AND ALL-POINT AP/ INFERENCE LATENCY FOR aiMOTIVE).

3) *Impact of Fusion Strategy*: We evaluate the effectiveness of the spatial-, channel-, and pixel-level attention along the dynamic adaptation mechanism in Table VIII, using aiMotive under both in-distribution (ID) and simulated OOD scenarios. The results demonstrate a consistent performance improvement as more fusion mechanisms are incorporated. When all components are combined, our approach improves all-point and 11-point AP by 7.53% and 7.07%, respectively, under OOD conditions, and by 7.58% and 7.36%, respectively, under the ID conditions over the baseline, validating its effectiveness in improving robustness and generalization.

V. CONCLUSION

We present CRUISE, a novel uncertainty-aware multi-modal sensor fusion framework that significantly improves the robust-

Feature Enhancement		Dynamic	All-point / 11-point AP (%)	
Spatial/Channel	Pixel	Adaptation	ID	OOD
✗	✗	✗	65.84 / 64.47	61.32 / 59.85
✓	✗	✗	67.53 / 66.05	62.74 / 61.12
✗	✓	✗	67.82 / 66.23	63.08 / 61.43
✓	✓	✗	69.31 / 67.64	64.25 / 62.63
✗	✗	✓	71.67 / 70.03	66.42 / 64.74
✓	✓	✓	73.42 / 71.83	68.85 / 66.92

TABLE VIII

IMPACT OF THE ATTENTION MECHANISMS AND DYNAMIC ADAPTATION MECHANISM IN THE FUSION MODULE ON AIMOTIVE

ness and generalization capabilities of AV perception systems. Our primary innovation lies in the effective combination of two key components: VLM-guided uncertainty estimation and a dynamic adaptation mechanism. The VLM module leverages extensive prior knowledge to transform coarse reliability scores into fine-grained, context-aware uncertainty heatmaps. The dynamic adaptation mechanism is specifically designed to explicitly model and capture cross-modal dependencies, allowing the framework to intelligently balance contributions from various sensors and compensate for uncertainty across modalities. Through these integrated innovations, CRUISE successfully mitigates uncertainty propagation and achieves superior performance in complex, OOD scenarios.

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